



Figure 1: YouTube Live Dashboard



Figure 2: Encoder Setup – Get the Server URL and Key

to go crazy at the uploading stage. Many video editors have YouTube presets (they also use FFmpeg as the backend!). For example, while rendering with OpenShot, take the profile as Web and target YouTube-HD.

Once you have your video file, sign into YouTube. You can now access the Live Streaming Dashboard using the link on the upload page or by visiting https://www.youtube.com/live_dashboard.

YouTube Live dashboard

Set things like the stream name and monetisation. At the bottom of the page, you can see ENCODER SETUP. Grab the server URL and stream name/key from there (by pressing the Reveal button).

Encoder setup: Get the server URL and key

Now open the terminal and use the following command:

```
ffmpeg -i <inputFilePath> -f flv SERVER_URL/KEY
```

Here is an example.

Server URL: `rtmp://a.rtmp.youtube.com/live2/`

Key: `nandakumar96.7mwj-vh6r-45ru-6eqq`

Command: `ffmpeg -i /tmp/a.flv -f flv rtmp://a.rtmp.youtube.com/live2/nandakumar96.7mwj-vh6r-45ru-6eqq`

Your stream is online! You can now use a few options like shown below:

```
-s widthxheight -- to scale the video
-b value -- to set the bitrate
```

These two options are useful if you want to degrade the

quality of the stream in case you have a low bandwidth. For example, you can stream an HD video file to 240p quality by using the following command:

```
ffmpeg -i /tmp/a.flv -f flv -s 426x240 -b 400k rtmp://...
```

Caution

Your live stream will be archived and uploaded to your channel once the broadcast is complete. To prevent this, you can edit the STREAM OPTIONS before starting the stream (in the live streaming dashboard).

Tip: A YouTube live stream is stopped when the encoder is terminated. You can press `Ctrl+C` to stop FFmpeg so that your stream also stops.

Make your screen public!

You've successfully broadcast a pre-recorded video. So why not try to screen cast it? Your screen is captured and broadcast live! The following command does that with your default microphone (or sound input device) as the audio source:

```
ffmpeg \
-f alsa -ac 2 -i default \
-f x11grab -framerate 24 -video_size 1366x768 \
-i :0.0+0,0 -preset medium -r 24 -g 48 \
-acodec libmp3lame -ar 44100 -threads 6 -qscale 3 -b:a
128k -bufsize 512k \
-f flv -s 426x240 SERVER_URL/KEY
```

The above command is split into several lines for readability. You may enter this in a single line avoiding the backslashes (`\`). If you use them, make sure you don't enter white spaces before entering the next line. Also, don't forget to press the `Enter` key after completing the command.

`video_size` tells you how much of the screen should be recorded. `-s` tells you what should be the output video resolution. In the above case, my whole screen (1366x768) is captured and scaled to 426x240. If your connection is fast enough, you can omit the second option so that a high quality video is uploaded (there's no problem if the viewer has a slow connection — YouTube will take care of that).

So happy streaming! 

References

- <https://support.google.com/youtube/answer/2797387>
- <https://support.google.com/youtube/answer/2907883>

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